Blackbody Radiation

PHYS 2203 — Introductory Physics III

October 16, 2026

Overview

With special relativity complete, we turn to the second revolution of early twentieth-century physics: quantum mechanics. It began with experiments on light that classical physics could not explain. This lecture covers the first of them, the spectrum of light emitted by a hot object. Classical theory, which gives every standing wave in a hot cavity the same average energy $k_B T$, predicts an infinite amount of radiation at high frequencies. Planck's solution to this problem is the subject of the next lecture.

1 Thermal Radiation

Every object emits electromagnetic radiation. The charges in a material are in constant thermal motion, and accelerating charges radiate. The amount and color of this **thermal radiation** depend on temperature. Coals in a fire glow red. An electric stove coil glows red when hot and brighter red as it gets hotter. At room temperature objects still radiate, but in the infrared, where our eyes cannot see it.

Most objects also *reflect* light, which complicates the picture. A **blackbody** is an object that absorbs all radiation striking it, so that everything it gives off is thermal radiation set by its temperature alone. The Sun's surface, which reflects almost nothing, is a good example.

A practical blackbody is a hollow object with a small hole in it (Figure 1). Radiation entering the hole bounces around inside and is absorbed long before it can find its way back out. Meanwhile the charges in the walls continually emit and absorb radiation, so the cavity fills with electromagnetic waves in equilibrium with the walls at temperature T. The radiation leaking out of the hole depends only on T: the *hole* is a blackbody.

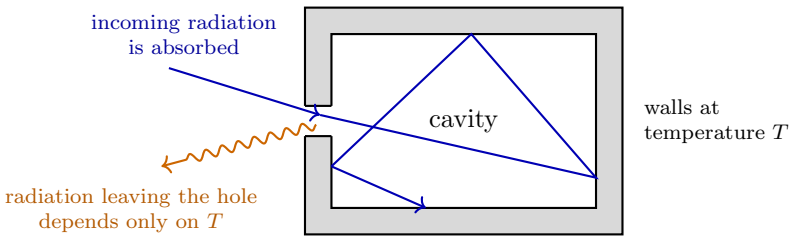


Figure 1: A cavity with a small hole behaves as a blackbody. Radiation entering the hole is absorbed after many reflections. The radiation leaving the hole is produced by the thermal motion of charges in the walls.

2 The Measured Spectrum

To describe how the energy in the cavity is spread over frequencies, define the **spectral energy density** $\mathrm{d}U/\mathrm{d}f$:

$$\mathrm{d}U = \text{energy in the cavity carried by waves with frequencies between } f \text{ and } f + \mathrm{d}f. \tag{1}$$

Measured curves for three temperatures are shown in Figure 2. Each is small at low frequency, rises to a maximum, and then falls back toward zero at high frequency.

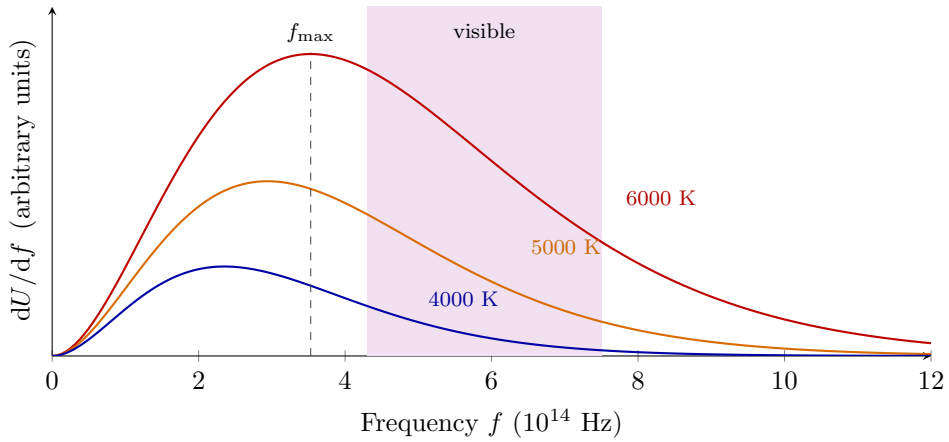


Figure 2: Measured blackbody spectra at three temperatures. As T increases the peak moves to higher frequency and the total area under the curve grows rapidly.

Two features of the measured curves were known by the 1890s.

Wien's displacement law. The frequency at which the spectrum peaks is proportional to the temperature:

$$\boxed{f_{\max} = \left(5.88 \times 10^{10}\ \mathrm{Hz/K}\right) T.} \tag{2}$$

The human body, at 310 K, peaks near 1.8×10^{13} Hz, in the infrared. This is the radiation detected by thermal cameras. A stove coil at around 1000 K peaks near 6×10^{13} Hz, also in the infrared, but the high-frequency tail of its spectrum reaches the red end of the visible spectrum, so it glows red. The Sun, at about 5800 K, peaks near 3.4×10^{14} Hz, just below the visible range of 4.3–7.5×10^{14} Hz, and a large part of its output is visible light.

Wien's law is often quoted in terms of wavelength instead, as $\lambda_{\max} T = 2.898 \times 10^{-3}$ m·K. That form describes the energy per unit *wavelength*, which peaks at a different place, so $\lambda_{\max}$ is not equal to $c/f_{\max}$.

Stefan's law. The total energy, the area under the curve, grows as the fourth power of temperature. The power emitted per unit area of a blackbody, its intensity, is

$$\boxed{I = \sigma T^4,} \qquad \sigma = 5.67 \times 10^{-8}\ \frac{\mathrm{W}}{\mathrm{m^2\,K^4}}. \tag{3}$$

Doubling the temperature increases the radiated power by a factor of $2^4 = 16$.

3 The Classical Prediction Fails: The Ultraviolet Catastrophe

Classical physics should be able to predict the shape of these curves. The radiation inside the cavity consists of electromagnetic **standing waves**, since the electric field must vanish at the conducting walls. In one dimension, a box of length L holds standing waves with wavelengths and frequencies

$$\lambda = \frac{2L}{n}, \qquad f = \frac{c}{\lambda} = \frac{nc}{2L}, \qquad n = 1, 2, 3, \ldots \tag{4}$$

as shown in Figure 3.

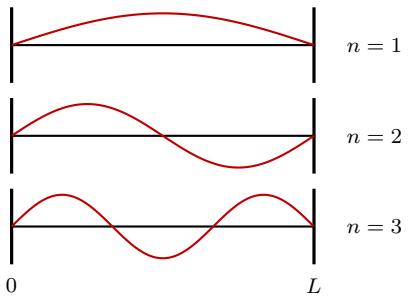


Figure 3: Standing waves in a one-dimensional box of length L. The field vanishes at both walls, so $\lambda = 2L/n$.

The classical prediction is the number of standing waves in each frequency range times the average energy of each one.

1. **Number of standing waves.** Many more high-frequency waves than low-frequency waves fit in the cavity. In a three-dimensional cavity of volume V, the number of standing waves with frequencies between f and $f + \mathrm{d}f$ is
$$\frac{8\pi V}{c^3} f^2 \, \mathrm{d}f. \tag{5}$$

2. **Average energy per wave.** In classical physics a wave can have any amplitude, and therefore any energy. When the waves share the thermal energy of the walls, classical physics predicts that each one ends up with the same average energy, whatever its frequency:
$$\bar{E}_{\text{classical}} = k_B T. \tag{6}$$
This follows from the **equipartition theorem** of classical physics: in thermal equilibrium, energy is shared equally among all the ways a system can store it, with an average of $\frac{1}{2}k_BT$ for each. A standing wave stores energy in its electric field and in its magnetic field, so it gets $2 \times \frac{1}{2}k_BT = k_BT$. The same theorem gives the average kinetic energy $\frac{3}{2}k_BT$ of a molecule in an ideal gas, with $\frac{1}{2}k_BT$ for motion along each of the three directions.

Boltzmann's constant.

$$k_B = 1.38 \times 10^{-23} \text{ J/K} = 8.62 \times 10^{-5} \text{ eV/K}.$$

The product k_BT is the typical thermal energy at temperature T. At room temperature, $k_BT \approx 0.025$ eV $\approx \frac{1}{40}$ eV.

Multiplying the two gives the classical prediction, known as the **Rayleigh–Jeans law**:

$$\boxed{\frac{\mathrm{d}U}{\mathrm{d}f} = k_B T \times \frac{8\pi V}{c^3} f^2} \qquad \text{(classical prediction).} \tag{7}$$

At low frequencies this agrees with experiment. At high frequencies it is badly wrong (Figure 4): instead of turning over, the parabola grows without limit as f increases, and the total energy, the area under the curve, is infinite. Every object at any temperature would radiate infinite power, mostly in the ultraviolet and beyond. This failure is known as the **ultraviolet catastrophe**. It does not come from a bad approximation. It follows directly from the assumption that a wave can carry any amount of energy.

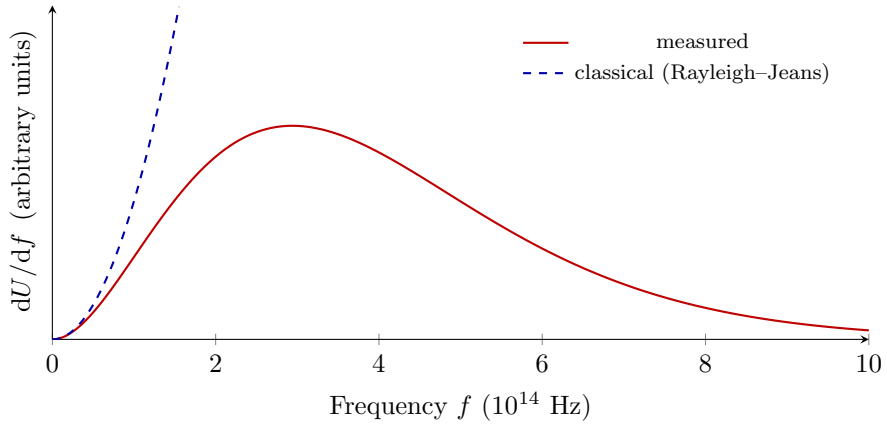


Figure 4: Measured spectrum at 5000 K (solid) and the classical prediction (dashed). They agree at low frequencies, but the classical curve rises without limit at high frequencies: the ultraviolet catastrophe.